

SARKER, ARANYA

Contact no : +8801721197243
Email : aranyasarker6@gmail.com
LinkedIn : www.linkedin.com/in/aranya-sarker-65ba67218/
Address : Dhaka, Bangladesh

CAREER OBJECTIVE

Aiming to utilize my knowledge of Analog, RF circuits and layouts and also having strong challenging mindsets, I would like to secure a position as a prospective graduate student in Analog Front End Design in USA.

Work Experience

Technical Expertise:

Layout Design	• Cadence Virtuoso Layout Editor (Tools: Layout-L, Layout-XL, PVS, Pegasus, Calibre, Assura) • Synopsys Custom Compiler (Tools: Custom Compiler Layout Editor, SDL, IC validator)
Schematic Design	• Cadence Virtuoso Layout Editor (Tools: Schematic Editor L, ADE-L, ADE Explorer, Spectre) • Synopsys Custom Compiler (Tools: Custom Compiler Schematic Editor, PrimeSim simulator, HSPICE)
Parasitic Extraction Tool	StartRC & QuantusRC(QRC),Assura
Technology	5nm,8nm, 12nm, 28nm, 40nm, 55nm, 90nm, 130nm.
Process	Planar MOSFET, FinFET, HVCMosfets.
TapeOut Experience	Explicitly worked on analog blocks right from schematic to GDS and have 9 tapeout experience.

Professional Project Works

Project 1:

Worked on **130nm** process node for PDK verification (DRC).

- Responsible for verify different cases to ensure the DRC manuals Information.
- **Challenges:** Verifying and making the right adjustment for every different cases for every layers existed on PDK.

Project 2 & 3:

Worked on **40nm & 130nm** process node with Analog to Digital Converter (ADC with HVCMosfets) layout.

- Responsible for floorplanning from block to top level layout and optimizing the area on the top level.
- Current mirrors, Input blocks all are done by the matching techniques.
- Ensured the symmetry of placements and routings for having same parasitic to parallel signals.
- **Specifications:** PDK: Infineon130nm, Bias currents to be updated from 500nA to 3uA, switching signals are in the 2.5v domain, Top block contains 5 subblocks (Timing block, en_dac_mux block, 1st stage, 2nd stage and feedforward sum & quantizer).
- **Challenges:** DRC, LVS, LatchUp, AntennaDRC and ERC issues, Sinkers connection, Bulk connections of different domains, Matching, Routing symmetry, HV mosfets placement to adjust blank space on the top, pin placement according to all subblocks and so on.

Project 4 & 5:

Worked on **5nm & 8nm** process node with Ring-Oscillator (FinFET based) that is used to profile the process:

- Responsible for designing and characterization of all devices.
- Did the pre-layout simulation and PVT analysis of block to Top level schematic.
- Responsible for floor-planning for the block and top level of layout.
- Worked on Parasitic extraction and post-layout simulation to ensure the timing constraints, signal integrity and confirming that the circuit functions correctly.
- Provided all the requirement files to the Client.
- **Specifications:** PDK: smsg05nm, Frequency: 1.2GHz, Inst Count: 132k (5nm), 101k(8nm), Number of Inverters: 32
- **Challenges:** DRC, LVS, Adjusting the parameters of the instances in the schematic to fulfil the specifications, Timing (All the flavors were completed within the deadline).

Project 6:

Worked on 28nm process node with Ring-Oscillator (Planar based) that is used to profile the process.

- Responsible for floor-planning for the block and top level of layout.

- Worked on Parasitic extraction and post-layout simulation to ensure the timing constraints, signal integrity and confirming that the circuit functions correctly.
- Provided all the requirement files to the Client.
- **Specifications:** PDK: TSMC28nm, Frequency: 1GHz, Inst Count: 72k, Area: 581[μm^2], Number of Inverters: 32
- **Challenges:** DRC, LVS, Adjusting the parameters of the instances in the schematic to fulfil the specifications, Timing (All the flavors were completed within the deadline).

Project 7:

Worked on 55nm process node with Level Shifter (Triple well FinFET).

- Completed layout of Level Shifter with Triple Well Mosfet.
- Routing was completed within metal 2.
- Optimized the area.
- Considered the width of the net of large amount of current.
- Used guard ring to separate all the blocks and for preventing internal block noise.
- **Specifications:** PDK: 55nm, Higher metal: Metal 2
- **Challenges:** DRC, LVS, Timing.

Project 8:

Worked on 12nm process node with BGR (FinFET based).

- Completed layout.
- Used common centroid matching for differential pair and interdigitation also the current mirror layout.
- Top level layout designed and verified.
- Used guard ring separately around the resistor and capacitor for blocking the internal noise.
- **Specifications:** PDK: GF12nm, Area: Must be Optimized.
- **Challenges:** DRC, LVS, Timing, BJT and their routing was a new challenge.

Project 9:

Worked on 90nm process node with Rail-to-Rail Op-Amp (Planer Mosfet).

- Completed layout.
- Used Matching technique.
- Considered sensitive nets and placed required device closed to the switch, maintained the width of the routing for passing different amount of current for long nets.
- **Specifications:** PDK: GF12nm, Area: Must be Optimized.
- **Challenges:** DRC, LVS, Timing.

ACADEMIC QUALIFICATION

Bachelor of Science in Electrical & Electronic Engineering (EEE) University

: American International University-Bangladesh (AIUB)

Academic Year : 2019-2022 (4th year)

Result : 3.92 out of 4

SPECIALIZATION

- | | |
|---|--|
| <ul style="list-style-type: none">• Computer Skills
Cadence
Synopsys | <ul style="list-style-type: none">• Communication skills Professional• Analog Circuit Skillful• Analog Layout Experienced |
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Special Qualification

- Quick learner & able to assume responsibility
- Experience in Outcome-based Education (OBE)
- Good sense of humor & know how to manage work with professional behavior
- Teamwork and leadership skills
- Having the ability to find the problem and quick solution

Awards

- Dean's award for academic performance (6 times)
For successfully earning a grade point average (GPA):
3.93 in Fall 19-20 semester,
3.95 in Spring 19-20 semester,
3.95 in Fall 20-21 semester,
3.93 in Spring 20-21 semester,
3.95 in Spring 21-22 semester,
4 in Fall 21-22 semester.
- Magna cum laude Award (For academic performance)

Personal Information

Father's Name : Bikash Chandra Sarker
Mother's Name : Lipika Rani Mondal
Date of Birth : 31-10-2000
Marital Status : Unmarried

Reference

1. Dr. Md. Abdul Mannan
American International University-Bangladesh
Program Director, Faculty of Engineering
Mobile : +8801799277730
Gmail : mdmannan@aiub.edu
2. Saadman Afzal
Neural Semiconductor Limited
Engineering Manager 1
Mobile : +8801717232405

- Gmail : saadman.afzal@neural-semiconductor.com
3. Dr. Jahid Hasan
Tyndall National Institute
Postdoctoral Researcher
Mobile : +353 83 088 9779
Gmail : jahid.hasan@tyndall.ie

I certify that, the information provided in this curriculum vitae is accurate and complete.

Aranya Sarker

Signature